

DAMMAM KING FAHD INTL, Saudi ArabiaWMO: **404150**Lat: **26.433N**Lon: **49.800E**Elev: **39**StdP: **14.68**Time Zone: **3.00 (E03)**Period: **99-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	44.5	46.7	17.7	13.5	64.7	21.5	16.2	66.7	29.9	60.8	25.2	59.0	8.9	280	0.530

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.9	115.2	70.0	113.3	69.7	111.4	69.6	83.8	93.9	82.0	93.5	80.2	93.1	12.5	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.0	161.2	91.1	78.9	150.2	90.8	76.8	139.6	90.2	48.2	94.1	46.2	93.4	44.1	92.9	91.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.9	24.0	21.2	DB	39.6	119.4	2.9	1.2	37.6	120.3	35.9	121.0	34.3	121.6	32.2	122.5	(4)
(5)			WB	33.1	85.8	3.2	1.6	30.8	87.0	29.0	87.9	27.2	88.8	24.9	89.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	81.3	60.3	63.6	71.3	80.9	91.1	96.4	99.0	98.1	92.7	84.9	72.5	63.4	(6)
(7)		DBStd	14.59	5.43	5.59	5.51	5.21	3.96	2.97	2.37	2.28	3.39	3.75	5.72	5.27	(7)
(8)		HDD50	2	2	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	360	166	84	9	0	0	0	0	0	0	0	8	93	(9)
(10)		CDD50	11421	321	381	661	926	1275	1392	1520	1492	1280	1082	676	415	(10)
(11)		CDD65	6302	19	44	205	476	810	942	1055	1027	830	617	234	43	(11)
(12)		CDH74	95113	127	347	1974	5546	12565	16086	18294	17345	12755	7800	2007	267	(12)
(13)		CDH80	64955	19	85	807	3006	8429	11795	13831	12885	8677	4617	769	35	(13)
(14)	Wind	WSAvg	9.7	9.7	10.5	10.3	10.1	10.4	11.8	10.5	8.6	8.2	7.6	8.8	9.6	(14)
(15)	Precipitation	PrecAvg	3.1	0.7	0.2	0.3	0.4	0.1	0.0	0.0	0.0	0.0	0.8	0.6	(15)	
(16)		PrecMax	4.9	2.2	1.1	0.8	1.1	0.9	0.0	0.0	0.0	0.0	3.9	3.0	(16)	
(17)		PrecMin	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)	
(18)		PrecStd	1.1	0.6	0.3	0.2	0.4	0.2	0.0	0.0	0.0	0.0	1.0	0.9	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.6	87.8	99.9	105.8	113.3	116.9	118.4	116.7	113.4	106.2	95.1	84.2	(19)
(20)			MCWB	59.5	61.7	62.5	65.8	68.3	69.5	70.3	71.1	69.5	67.3	64.9	62.9	(20)
(21)		2%	DB	77.2	82.2	93.3	101.1	109.7	114.5	115.5	114.6	109.8	103.6	91.5	79.7	(21)
(22)			MCWB	58.6	59.0	61.6	64.9	67.4	68.7	70.2	71.0	69.5	67.1	64.8	60.9	(22)
(23)		5%	DB	73.7	77.9	88.1	97.2	107.3	111.6	113.2	112.7	107.8	100.5	87.9	77.0	(23)
(24)			MCWB	58.0	57.8	60.4	64.0	66.9	68.0	69.8	71.0	69.3	67.1	64.7	61.2	(24)
(25)		10%	DB	70.8	74.8	84.0	94.0	104.4	109.5	111.2	110.8	105.8	97.8	84.4	73.6	(25)
(26)			MCWB	56.8	57.3	59.2	63.7	66.5	67.3	69.5	71.2	69.6	67.6	65.0	60.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.7	65.3	68.3	71.1	75.5	79.1	85.4	87.1	83.2	79.2	73.4	69.3	(27)
(28)			MCD B	72.1	74.4	79.7	84.3	91.1	91.3	93.8	94.6	93.1	87.9	81.9	75.0	(28)
(29)		2%	WB	63.4	63.4	66.4	69.2	73.0	75.6	83.1	84.5	81.2	77.2	71.3	66.8	(29)
(30)			MCD B	70.7	73.1	78.8	85.7	90.7	92.2	93.3	94.5	92.1	87.3	80.9	73.1	(30)
(31)		5%	WB	61.3	61.8	64.8	67.9	71.4	72.5	80.9	82.8	79.5	75.6	69.6	64.7	(31)
(32)			MCD B	69.5	71.8	77.7	85.0	90.8	96.5	93.6	94.0	91.5	85.9	79.4	71.9	(32)
(33)		10%	WB	59.1	60.3	63.0	66.7	69.7	70.2	78.1	81.2	77.7	74.0	67.9	62.6	(33)
(34)			MCD B	68.0	70.5	77.5	85.0	92.9	100.9	95.4	93.6	91.1	85.3	78.8	71.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	19.6	20.7	23.5	23.8	26.1	26.7	25.9	26.5	27.5	26.6	20.9	19.3	(35)
(36)			MCDBR	23.2	25.4	30.7	29.9	30.2	30.2	29.1	29.2	29.6	29.6	25.4	22.9	(36)
(37)			MCWBR	10.3	9.6	10.2	8.5	9.5	10.7	11.8	11.7	12.1	11.2	10.3	10.2	(37)
(38)		5% WB	MCD BR	17.9	19.4	21.7	22.9	26.4	28.9	26.8	25.0	26.2	25.7	21.2	16.8	(38)
(39)			MCWBR	9.7	8.9	9.3	7.9	9.6	11.7	12.8	11.8	12.4	10.7	10.2	8.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.525	0.579	0.640	0.703	0.701	0.673	0.694	0.641	0.571	0.534	0.540	0.506	(40)	
(41)		taud	1.783	1.671	1.552	1.486	1.482	1.515	1.518	1.616	1.730	1.805	1.792	1.838	(41)	
(42)		Ebn at Noon	221	221	217	209	209	214	209	220	231	230	215	220	(42)	
(43)		Edh at Noon	61	74	87	95	96	92	92	83	72	63	60	56	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1206	1443	1728	1923	2223	2401	2283	2175	1987	1655	1241	1131	(44)	
(45)		RadStd	81	87	124	112	65	67	51	37	36	49	89	82	45	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.50	N/A	N/A	+1.35	N/A	N/A	N/A	N/A	+205	N/A
(47)	Regional (3 neighbors)	+0.85	N/A	-3.29	+1.35	N/A	N/A	N/A	N/A	+306	+283

Nomenclature: See separate page